Paper 2: Structure of the marking scheme

Candidate responses are marked according to different scales, depending on the types of response anticipated. Scales labelled A divide candidate responses into two categories (correct and incorrect). Scales labelled B divide responses into three categories (correct, partially correct, and incorrect), and so on. The scales and the marks that they generate are summarised in this table:

Scale label	A	B	C	D
No of categories	2	3	4	5
5-mark scale	0, 5	0, 2, 5	0, 2, 3, 5	
10-mark scale			0, 4, 6, 10	0, 3, 5, 7, 10
15-mark scale			0, 6, 8, 15	0, 4, 6, 8, 15

A general descriptor of each point on each scale is given below. More specific directions in relation to interpreting the scales in the context of each question are given in the scheme, where necessary.

Marking scales – level descriptors

A-scales (two categories)
- response of no substantial merit (no credit)
- correct response (full credit)

B-scales (three categories)
- response of no substantial merit (no credit)
- partially correct response (partial credit)
- correct response (full credit)

C-scales (four categories)
- response of no substantial merit (no credit)
- response with some merit (low partial credit)
- almost correct response (high partial credit)
- correct response (full credit)

D-scales (five categories)
- response of no substantial merit (no credit)
- response with some merit (low partial credit)
- response about half-right (mid partial credit)
- almost correct response (high partial credit)
- correct response (full credit)

In certain cases, typically involving incorrect rounding, omission of units, a misreading that does not oversimplify the work, or an arithmetical error that does not oversimplify the work, a mark that is one mark below the full-credit mark may also be awarded. Such cases are denoted with a * and this level of credit is referred to as *Full Credit -1*. Thus, for example, in Scale 10C, *Full Credit -1* of 9 marks may be awarded.

The only marks that may be awarded for a question are those on the scale above, or *Full Credit -1*.

In general, accept a candidate's work in one part of a question for use in subsequent parts of the question, unless this oversimplifies the work involved.

In general, an answer without sufficient supporting work is awarded the highest level of partial credit below *Full Credit –1* (typically *Partial Credit* or *High Partial Credit*, as appropriate).

Paper 2: Summary of mark allocations and scales to be applied

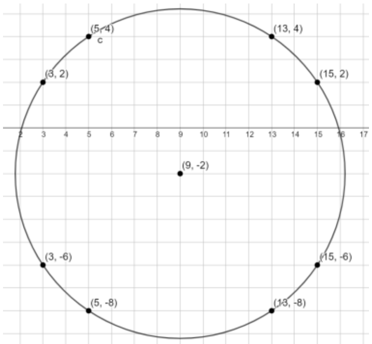
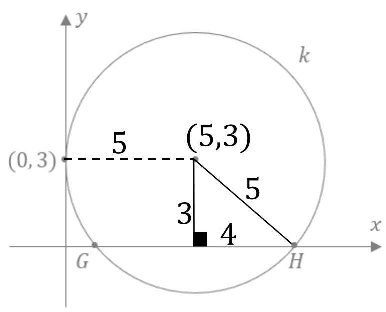
Section A (150) **Answer any five questions**		**Section B (150)** **Answer any three questions**	
Question 1 (30) (a)(i)(ii) 15D (a)(iii) 5C (b) 10D **Question 2 (30)** (a) 10C (b) 10C (c) 10D **Question 3 (30)** (a) 10C (b) 10D (c) 10D	**Question 4 (30)** (a) 10C (b) 10C (c) 10D **Question 5 (30)** (a)(i)(ii) 10D (b)(i) 5B (b)(ii) 5B (b)(iii) 10C **Question 6 (30)** (a)(i) 5B (a)(ii) 10C (b) 15C	**Question 7 (50)** (a)(i) 10D (a) (ii) 10D (b) 10D (c)(i)(ii) 10D (d)(i)(ii) 10C **Question 8 (50)** (a)(i) 5C (a)(ii) 10C (b) 5C (c) 10C (d) 10C (e) 10D	**Question 9 (50)** (a)(i) 5B (a)(ii) 10C (b)(i)(ii) 10C (c) 15D (d)(i)(ii) 10D **Question 10 (50)** (a) 15D (b) 5A (c)(i) 5C (c)(ii) 10C (d)(i) 5C (d)(ii) 10D

Paper 2: Model Solutions & Marking Notes

Note: The model solutions for each question are not intended to be exhaustive – there may be other correct solutions.

Q1	Model Solution – 30 Marks	Marking Notes
(a) (i) (ii)	(i) $P(6,5)\ \ Q(4,-3)$ $$M=\left(\frac{6+4}{2},\frac{5+(-3)}{2}\right)=(5,1)$$ (ii) $m_{PQ}=\frac{(-3)-(5)}{(4)-(6)}=4$ $m_{bisector}=-\frac{1}{4}$ $y-y_1=m(x-x_1)$ $y-1=-\frac{1}{4}(x-5)$ $4y-4=-x+5$ $x+4y-9=0$	**Scale 15D (0, 4, 6, 8, 15)** **4 steps:** **1.** Find the midpoint of PQ **2.** $Find\ the\ slope\ of\ PQ$ **3.** Find slope of perpendicular bisector of PQ **4.** Find equation of perpendicular bisector *Low Partial Credit* • Work of merit, for example, midpoint formula with some correct substitution; slope formula with some correct substitution *Mid Partial Credit* • 1 steps correct *High Partial Credit* • 3 steps correct
(a) (iii)	Perpendicular bisector of [QR]: $x=7$ $x+4y-9=0$ $7+4y-9=0$ $4y=2$ $y=\frac{1}{2}$ $(7,\frac{1}{2})$	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit* • Work of merit, for example, effort to calculate perpendicular bisector of [PR] or [QR] *High Partial Credit* • $x=7$ • $y=\frac{1}{2}$

Q1	Model Solution – 30 Marks	Marking Notes
(b)	Let $A=(x,0)$ and $B=(0,y)$ $Midpoint=\left(\frac{x+0}{2},\frac{0+(y)}{2}\right)$ $(-6,2)=\left(\frac{x}{2},\frac{y}{2}\right)$ $\frac{x}{2}=-6$ $\qquad$ $\frac{y}{2}=2$ $x=-12$ $\qquad$ $y=4$ $A=(-12,0)$ $\qquad$ $B=(0,4)$	**Scale 10D (0, 3, 5, 7, 10)** *Low Partial Credit* • Work of merit, for example, states that $y=0$ in point A or that $x=0$ in point *Mid Partial Credit* • Correct substitution into midpoint formula • $A=(x,0)$ and $B=(0,y)$ *High Partial Credit* • $\boldsymbol{\frac{x}{2}=-6}$ **and** $\boldsymbol{\frac{y}{2}=2}$ • $\boldsymbol{x}$ correct or $\boldsymbol{y}$ correct

Q2	Model Solution – 30 Marks	Marking Notes
(a)	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - x_1)^2}$ $r = \sqrt{(-1-2)^2 + (-3-4)^2}$ $r = \sqrt{9+49} = \sqrt{58}$ $(x-2)^2 + (y-4)^2 = 58$ *or* $x^2 + y^2 - 4x - 8y - 38 = 0$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Work of merit in calculating the radius • Subs $(2, 4)$ into the equations of a circle formula *High Partial Credit* • Calculates the radius of the circle • Calculates r^2
(b)	 Any 2 of the following points: $(15, 2)$, $(15, -6)$, $(5, 4)$, $(5, -8)$, $(3, 2)$, $(13, 4)$, *or* $(13, -8)$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Work of merit, for example, some correct transformation *High Partial Credit* • One correct point • Finds the equation of the circle
(c)	 $r = \sqrt{3^2 + 4^2} = 5$ Centre: $(5, 3)$ Equation of Circle: $(x-5)^2 + (y-3)^2 = 25$	**Scale 10D (0, 3, 5, 7, 10)** *Low Partial Credit* • Work of merit, for example, calculates 3 as the vertical height of the centre above the x-axis • Shows that [GH] is bisected by the perpendicular *Mid Partial Credit* • Finds the radius of the circle *High Partial Credit* • Finds centre of the circle

Q3	Model Solution – 30 Marks	Marking Notes
(a)	$Tangent\ correctly\ drawn,$ $with\ construction\ lines$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Work of merit, for example, some indication of understanding that tangent $\perp$ radius • Perpendicular line drawn without construction lines *High Partial Credit* • Line PQ with 2 points equidistant from Q marked on the line
(b)	$l = 2\pi r$ $30 = 2\pi r$ $\frac{15}{\pi} = r$ $A = \pi r^2$ $A = \pi \left(\frac{15}{\pi}\right)^2$ $A = \frac{225}{\pi}\ cm^2$	**Scale 10D (0, 3, 5, 7, 10)** *Low Partial Credit* • Work of merit, for example, $2\pi r = 30$ *Mid Partial Credit* • Finds the radius in terms of π *High Partial Credit* • Correct substitution into the area of a circle formula *Full Credit -1* • Answer not in correct form
(c)	$\lvert\angle BCO\rvert = x$ $so\ \lvert\angle BOC\rvert = x$ [isosceles Δ as $\lvert BC\rvert = \lvert BO\rvert$] $so\ \lvert\angle CBO\rvert = 180 - 2x$ [angles in a Δ] $so\ \lvert\angle ABO\rvert = 180 - (180 - 2x) = 2x$ [straight angle] $so\ \lvert\angle BAO\rvert = 2x$ [isosceles Δ as as $\lvert BO\rvert = \lvert AO\rvert$] $so\ \lvert\angle BOA\rvert = 180 - 2x - 2x = 180 - 4x$ [angles in a Δ] $so\lvert\angle AOB\rvert = 180 - (180 - 4x) - (x) = 3x$ [straight angle]	**Scale 10D (0, 3, 5, 7, 10)** *Low Partial Credit* • Work of merit, for example, marks isosceles triangles or finds $\lvert\angle BOC\rvert$ in terms of x *Mid Partial Credit* • Finds $\lvert\angle ABO\rvert$ in terms of x *High Partial Credit* • Finds $\lvert\angle BAO\rvert$ in terms of x

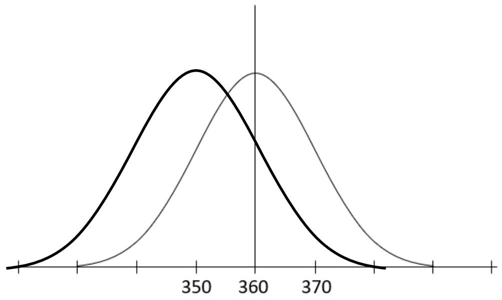
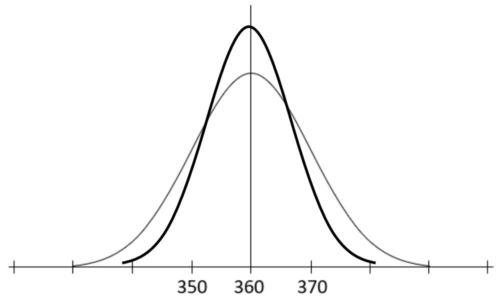
Q4	Model Solution – 30 Marks	Marking Notes
(a)	$\text{Cos}(A+B) = CosACosB - SinASinB$ $\text{Cos}(105) = \text{C}os(60+45)$ $\text{Cos}(60+45) = Cos60Cos45 - Sin60Sin45$ $\text{Cos}(105) = \left(\frac{1}{2}\right)\left(\frac{1}{\sqrt{2}}\right) - \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{\sqrt{2}}\right)$ $\text{Cos}(105) = \frac{1-\sqrt{3}}{2\sqrt{2}}$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Work of merit, for example, $60+45$ or some correct substitution into relevant formula *High Partial Credit* • $Cos60Cos45 - Sin60Sin45$ or equivalent
(b)	$Sin(A+B) = SinACosB + CosASinB$ Replace B with A $Sin(A+A) = SinACosA + CosASin$ $Sin2A = 2SinACosB$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Work of merit, for example, $Sin(A+B)$ formula *High Partial Credit* • $Sin(A+A) = SinACosA + CosASinA$
(c)	$\frac{SinW}{3} = \frac{Sin2W}{5}$ $5SinW = 3Sin2W$ $5SinW = 3[2SinWCosW]$ $5 = 6CosW$ $CosW = \frac{5}{6}$	**Scale 10D (0, 3, 5, 7, 10)** *Low Partial Credit* • Work of merit, for example, some correct substitution into the sine rule *Mid Partial Credit* • Fully correct substitution into the Sine rule *High Partial Credit* • $5SinW = 3[2SinWCosW]$

Q5	Model Solution – 30 Marks	Marking Notes
(a) **(i)** **(ii)**	$(i)\,\frac{1}{\sqrt{1600}} = \frac{1}{40} = 0.025 = 2.5\%$ (ii) $\frac{1}{\sqrt{n}} = 0.01$ $1 = 0.01\sqrt{n}$ $\frac{1}{0.01} = \sqrt{n}$ $100 = \sqrt{n}$ $n = 10000$	**Scale 10D (0, 3, 5, 7, 10)** *Low Partial Credit* • Work of merit, for example, substitutes 1600 into the formula *Mid Partial Credit* • One part correct • Work of merit in both parts *High Partial Credit* • One part correct and work of merit in the other
(b) **(i)**	$P(A \cap B) = P(A).P(B)$ $P(A \cap B) = 0.2 \times 0.3 = 0.06$	**Scale 5B (0, 2, 5)** • Some work of merit, for example, correct formula
(b) **(ii)**	$P(C \cup D) = P(C) + P(D) - P(C \cap D)$ $P(C \cup D) = 0.4 + 0.5 - 0.08 = 0.82$	**Scale 5B (0, 2, 5)** • Some work of merit, for example $\boldsymbol{P(C) + P(D)}$; Correct Venn diagram
(b) **(iii)**	$P(E\|F) = \frac{P(E \cap F)}{P(F)}$ $\frac{1}{4} = \frac{x}{x+0.4}$ $x + 0.4 = 4x$ $0.4 = 3x$ $x = \frac{0.4}{3} = \frac{2}{15} = 0.1\dot{3}$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Work of merit, for example, correct formula written *High Partial Credit* • $\frac{1}{4} = \frac{x}{x+0.4}$

Q6	Model Solution – 30 Marks	Marking Notes
(a) (i)	$7! = 5040$	**Scale 5B (0, 2, 5)** • Work of merit, for example, some arrangements shown
(a) (ii)	(AE) together: (AE),B,C,D,F,G $6! \times 2! = 720 \times 2 = 1440$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit:* • Work of merit, for example, one arrangement shown *High Partial Credit:* • 6! calculated
(b)	2,2,5,6,7,8 $[\text{Mean} = \frac{30}{6} = 5]$	**Scale 15C (0, 6, 8, 15)** *Low Partial Credit:* • Work of merit, for example, has a solution where 2 is the mode **or** 5.5 is the median *High Partial Credit:* • Solution where 2 is the mode **and** 5.5 is the median, but mean > 5

Q7	Model Solution – 50 Marks	Marking Notes
(a) (i)	$\frac{330-360}{27} = -1.11$ $\frac{390-360}{27} = 1.11$ $P(330 < X_{mean} < 390)$ $= P(-1.11 < z < 1.11)$ $= 0.8665 - (1 - 0.8665)$ $= 0.8665 - 0.1335$ $= 0.733$	**Scale 10D (0, 3, 5, 7, 10)** **4 Parts:** **1.** Finds z-score for 330 **2.** Finds z-score for 390 **3.** Finds 0.8665 **4.** Finishes *Low Partial Credit* • Work of merit, for example draws relevant diagram, or indicates μ or σ *Mid Partial Credit* • 2 steps correct *High Partial Credit* • 3 steps correct
(a) (ii)	$p(z < z_1) = 0.95$ $z_1 = 1.64$ $\therefore z = -1.64$ $\frac{x-360}{27} = -1.64$ $x - 360 = -44.28$ $x = 315.72\ cm$	**Scale 10D (0, 3, 5, 7, 10)** **4 steps:** **1.** Finds z-score of 1.64 **2.** Finds z-score of -1.64 **3.** Subs value into the z-score formula **4.** Finds x *Low Partial Credit* • Work of merit, for example, draws a relevant diagram or mentions 95%/0.95 *Mid Partial Credit* • 2 steps correct *High Partial Credit* • 3 steps correct *Full Credit-1* • No unit or incorrect unit

Q7	Model Solution – 50 Marks	Marking Notes
(b)	***Alternative Hypothesis:*** $The\ mean\ height\ of\ the\ trees\ in\ Fiadh's\ forest\ is\ not\ 360cm$ **Calculations:** $$\boldsymbol{z} = \frac{350-360}{\frac{27}{\sqrt{100}}} = -3.7$$ **Conclusion:** $We\ reject\ the\ the\ null\ hypothesis$ **Reason for your Conclusion:** $-3.7 < -1.96\ so\ the\ test\ statistic\ is\ in\ the\ critical\ zone\ of\ rejection$	**Scale 10D (0, 3, 5, 7, 10)** **4 steps:** **1.** States alternative hypothesis **2.** Calculations **3.** Conclusion **4.** Reason *Low Partial Credit* • Work of merit *Mid Partial Credit* • 2 steps correct *High Partial Credit* • 3 steps correct
(c) **(i)** **(ii)**	$\hat{p} = \frac{48}{160} = 0.3$ $E = 1.96\sqrt{\frac{0.3(1-0.3)}{160}} = 0.071$ $\hat{p} - E < p < \hat{p} + E$ $0.3 - 0.071 < p < 0.3 + 0.071$ $0.229 < p < 0.371$ $In\ 100\ samples, you\ would\ expect\ 95\ of\ them\ to\ have\ a\ mean\ within\ this\ interval$ j	**Scale 10D (0, 3, 5, 7, 10)** **4 steps:** **1.** Calculate $\hat{p}$ **2.** Calculate E **3.** Find Interval **4.** Explanation *Low Partial Credit* • Work of merit *Mid Partial Credit* • 2 steps correct *High Partial Credit* • 3 steps correct

Q7	Model Solution – 50 Marks	Marking Notes
(d) **(i)** **(ii)**	*A graph that has been validly transformed, for example:*  	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Work of merit, for example, any horizontal movement in (i) or vertical movement in (ii). *High Partial Credit* • One graph correct • Work of merit in both parts

Q8	Model Solution – 50 Marks	Marking Notes
(a) **(i)**	$A = \frac{1}{2}abSinC$ $A = \frac{1}{2}(59)(47)Sin28$ $A = 650.9 \ \approx 651\ [m^2]$	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit* • Work of merit, for example some correct substitution into the formula *High Partial Credit* • Fully correct substitution into formula
(a) **(ii)**	$a^2 = b^2 + c^2 - 2bcCosA$ $a^2 = 59^2 + 47^2 - 2(59)(47)Cos28$ $a^2 = 793.17$ $a = 28.16\ m$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit:* • Work of merit, for example, some correct substitution into the cosine formula *High Partial Credit:* • Fully substituted Cosine Rule formula *Full Credit-1* • Incorrect units or no units • Incorrect rounding or no rounding • Calculator in incorrect mode
(b)	$time = \frac{distance}{speed} = \frac{4.8\ km}{8\ km/hr} = 0.6\ hours$ $time = 0.6 \times 60 = 36\ [minutes]$	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit* • Work of merit, for example, $t = \frac{d}{s}$ *High Partial Credit* • Calculates 0.6 hours

Q8	Model Solution – 50 Marks	Marking Notes
(c)	$Volume = l \times w \times h$ $0.415 = (5x)(3x)(2x)$ $0.415 = 30x^3$ $0.0138\dot{3} = x^3$ $x = \sqrt[3]{0.0138\dot{3}}$ $x = 0.24\ [m]$ Height $= 0.24 \times 2 = 0.48\ m = 48\ [cm]$ Width$= 0.24 \times 3 = 0.72\ m = 72\ [cm]$ Height$= 0.24 \times 5 = 1.2\ m = 120\ [cm]$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Work of merit, for example, lets 0.415 = $l \times w \times h$ *High Partial Credit* • Substantial work of merit • Calculates x
(d)	$height = 2r$ $V = \pi r^2 h$ $2.78 = \pi r^2(2r)$ $0.44245 = r^3$ $\sqrt[3]{0.44245} = r$ $0.762 = r$ $r = 0.76\ m$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Work of merit, for example, identifies h as $2r$ *High Partial Credit* • Fully correct substitution into formula [3rd line of solution]
(e)	length of Small Square = $r \times 2 = 2\text{cm}$ diagonal of small square = $\sqrt{2^2 + 2^2} = 2\sqrt{2}$ Diameter of big circle $= 2\sqrt{2} + 2$ Length of Big Square $= 2\sqrt{2} + 2$ Area of Big Square = $\left(2\sqrt{2} + 2\right)^2$ $= 12 + 8\sqrt{2}\ cm^2$	**Scale 10D (0, 3, 5, 7, 10)** *Low Partial Credit* • Work of merit, for example, finds length of small square *Mid Partial Credit* • Finds diagonal of small square *High Partial Credit* • Finds radius or diameter of big circle

Q9	Model Solution – 50 Marks	Marking Notes
(a) (i)	0.9 There is a strong positive correlation	**Scale 5B (0, 2, 5)** *Partial Credit* • Work of merit in justification • Correct box ticked
(a) (ii)	Reasonable line of best fit drawn on the graph 90 kg [Or their correct value]	**Scale 10C (0, 4, 6, 10)** Note: Accept line of best fit drawn with some values on each side and with reasonable slope *Low Partial Credit* • Work of merit, for example, line drawn with positive slope *High Partial Credit* • One part correct
(b) (i) (ii)	$0.92^3 \times 0.08 = 0.06229504$ $= 0.0623$ (ii) $1 - \binom{20}{0} 0.92^{20} - \binom{20}{1} 0.92^{19} 0.08^1$ $= 1 - 0.1886933292 - 0.3281623116$ $= 0.4831$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit* • Work of merit, for example, mentions 0.92 *High Partial Credit* • One part correct • Work of merit in both parts
kll(c)	$x = 4y$ .....$Eq. A$ $x(7) + y(5) + z(0) = 1.155$....$Eq. B$ $x + y + z = 1$....$Eq. C$ $4y(7) + y(5) + z(0) = 1.155$ $28y + 5y = 1.155$ $33y = 1.155$ $y = 0.035$ $x = 4(0.035) = 0.14$ $z = 1 - 0.035 - 0.14 = 0.825$	**Scale 15D (0, 4, 6, 8, 15)** *Low Partial Credit* • Work of merit, for example, lets $x = 4y$ *Mid Partial Credit* • Two of $\boldsymbol{A}, \boldsymbol{B}$ *or* $\boldsymbol{C}$ *High Partial Credit* • Finds x or y

Q9	Model Solution – 50 Marks	Marking Notes
(d) **(i)**	$7\ C_3 = 35$	**Scale 10D (0, 3, 5, 7, 10)** *Low Partial Credit* • Work of merit, for example, *Mid Partial Credit* • One part correct • Work of merit in both parts *High Partial Credit* • One part correct and work of merit in the other
(ii)	$\frac{4C_3}{7C_3} = \frac{4}{35}$	

<table>
<tr><th>Q10</th><th>Model Solution – 50 Marks</th><th>Marking Notes</th></tr>
<tr><td>(a)
(i)(ii)</td><td><table><tr><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>0</td><td>30</td><td>60</td><td>30</td><td>0</td><td>30</td><td>60</td><td>30</td><td>0</td></tr></table>
Correct Graph Drawn</td><td>Scale 15D (0, 4, 6, 8, 15)
Solution consists of 18 parts:
-8 values in table
-9 points plotted
-Points joined appropriately

Low Partial Credit
• Work of merit in finding one value in table
• 1 part correct

Mid Partial Credit
• 8 parts correct

High Partial Credit
• 15 parts correct

Full Credit -1
• 17 parts correct
• Correct graph without table</td></tr>
<tr><td>(b)</td><td>It takes 0.4 seconds for the wheel to do 1 full rotation</td><td>Scale 5A (0, 5)</td></tr>
<tr><td>(c)
(i)</td><td>$l = 2\pi r$
$l = 2\pi(300mm)$
$l = 1884.955592$
$l \approx 1885mm$</td><td>Scale 5C (0, 2, 3, 5)

Low Partial Credit
• Work of merit, for example, $r = 300mm$

High Partial Credit
• Correct substitution into the formula</td></tr>
<tr><td>(c)
(ii)</td><td>$s = \frac{d}{t} = \frac{1885}{0.4} = \frac{9425}{2}\ mm/sec$
$= 282750\ mm/min$
$= 16965000\ mm/hr$
$= 16.965\ km/hr$
$= 17\ [km/hr]$</td><td>Scale 10C (0, 4, 6, 10)

Low Partial Credit
• Work of merit, for example $s = \frac{d}{t}$

High Partial Credit
• Some correct speed calculated but in incorrect unit</td></tr>
</table>

Q10	Model Solution – 50 Marks	Marking Notes
(d) **(i)**	$\tan(30) = \frac{h}{\lvert OB \rvert}$ $\lvert OB \rvert = \frac{h}{\tan(30)} = \sqrt{3}\,h$	**Scale 5C (0, 2, 3, 5))** *Low Partial Credit* • Work of merit, for example, some correct substitution into trigonometric formula *High Partial Credit* • Correct trigonometric equation
(d) **(ii)**	$a^2 = b^2 + c^2 - 2bcCosA$ $120^2 = h^2 + (\sqrt{3}h)^2 - 2(h)\left(\sqrt{3}h\right)Cos60$ $14400 = h^2 + 3h^2 - \sqrt{3}h^2$ $14400 = h^2(4 - \sqrt{3})$ $\frac{14400}{(4-\sqrt{3})} = h^2$ $h = \sqrt{\frac{14400}{(4-\sqrt{3})}} = 79.7$ [m]	**Scale 10D (0, 3, 5, 7, 10)** *Low Partial Credit* • Work of merit, for example, some correct substitution into the cosine rule *Mid Partial Credit* • Fully correct substitution into the cosine rule *High Partial Credit* • Isolates h^2